

Fine Tuning the Optical Properties of Single Au Nanoparticles by Plasmon-Driven Growth in Closed-Loop Control

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The morphology and composition of metallic nanoparticles (NPs) determine their optical response, so finely controlling them at the single particle level is the key to achieve tailored functionalities. Here, the control of plasmon-driven growth of Au NPs is studied by live monitoring their photoluminescence emission in a closed-loop. It is found that the final emission maximum of single NPs can be tuned between 550 and 580 nm with a precision of 2–3 nm, and that the tuning is also reflected in their scattering maximum. In comparison to controlling the growth by irradiation time and/or reaction conditions, the closed-loop control delivers superior reproducibility and a three to fourfold higher precision in the final properties of the NPs.

1. Introduction

Metallic nanoparticles (NPs) present unique optical properties due to their localized surface plasmon resonances that enable the nanoscale manipulation of optical fields, forces, and heat.^[1] These features have put them in the spotlight for the last couple of decades, in search of new materials for photonics,^[2] microscopy,^[3] plasmon-driven chemical reactions,^[4–6] solar-fuel generation,^[7,8] ultra-sensitive sensing,^[9,10] and medicine,^[11] among other applications. Localized surface plasmon resonances are determined by the shape, size, and material composition of the NPs.^[12] Still, the fabrication of supported arrangements of NPs where each individual element has tailored properties remains a bottleneck in nanotechnology.^[13]

Top-down techniques such as electron beam lithography allow high-resolution fabrication of supported NPs in arbitrary

patterns. However, they are limited in terms of combining materials, the typical crystallinity of the materials is usually poor, and 3D geometries such as spheres, nanostars, or core-shells are impossible or impractical to obtain. Electrochemical growth of NPs allows the fabrication of 3D geometries such as nanoflowers, but with limited possibilities of pattern design, material combination, and choice of substrate.^[14–16]

On the other hand, colloidal chemistry offers enormous versatility to obtain NPs with different shapes, compositions, and sizes by modulating the synthesis conditions.^[17,18]

Plasmon-driven reactions use light as an additional degree of freedom to tune the final morphology and size of colloidal metallic NPs.^[19] For example, plasmon-driven reduction of Ag⁺ has been used to produce colloidal Ag NPs of various shapes from Ag spherical seeds,^[20,21] as well as Au@Ag core-shell NPs from Au seeds.^[22] Then, the challenge of how to bring the colloidal NPs from the liquid phase to specific positions of a substrate remains.

The simplest way to attach colloidal NPs to substrates is through electrostatic deposition, but obtaining ordered arrays is not possible this way.^[23] Ordered arrays of NPs can be obtained by self-assembly methods, but the versatility is limited to their thermodynamically most stable configurations.^[24] Template-assisted self-assembly adds flexibility of pattern design but requires additional steps than lengthen and complicate the fabrication procedure.^[25,26] Maximum flexibility can be obtained by methods that utilize an AFM tip^[27] or focused laser light^[28] to drive deposition in specific regions of the substrate. In general, combining different NPs with all these methods remains a challenge.

An emerging method to deposit colloidal NPs in predefined positions of a substrate with high accuracy and flexibility of pattern design is optical printing.^[29–34] The technique is based on optical forces and allows the printing of virtually any kind of NP that interacts with light. This technique is also able to selectively print subpopulations of NPs out of a polydisperse suspension.^[33,35]

An appealing route for tailoring the optical properties of individual supported NPs is to perform controlled plasmon-driven growth reactions on them. For example, the reduction of silver ions on Ag NPs,^[36] the reduction of aqueous PtCl₄^{2–} on Au NPs,^[37] and the copper oxide reduction on Cu NPs^[38] were demonstrated by this approach. Also, supported Au nanoprisms,

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nanospheres, and nanodiscs^[39–42] were obtained by plasmon-driven reduction of aqueous HAuCl₄. In addition, anisotropic growth of Au NPs was achieved using polarized light.^[42] Typically, the final morphology of the NPs has been controlled by the irradiation intensity and time, and other reaction conditions (e.g., precursor concentrations). The final properties of the NPs have been usually analyzed after the reaction has been stopped, by dark-field spectroscopy and/or electron microscopy. Alternatively, photoluminescence (PL) spectroscopy was also employed for real time monitoring of Au nanorods growth^[43] and the fabrication of semiconductor shells around individual Au cores.^[44] This characterization method is not invasive if the same laser used for the plasmon-driven reaction is used to excite the PL. The PL spectra of Au NPs is sensitive to changes in the NPs shape,^[45,46] size,^[47] surface,^[48] and chemical environment,^[49] and therefore PL spectroscopy holds great potential for live monitoring of plasmon-driven reactions.

Here, we show that the optical properties of Au NPs can be finely tuned by plasmon-driven growth reactions controlled with a feedback loop analyzing the PL signal in real time. Single Au NPs, 66 nm in diameter, were first deposited onto glass substrates through optical printing and then grown by plasmon-driven reduction of aqueous HAuCl₄. By automating

the irradiation and the PL acquisition and analysis, it was possible to stop the reaction at predefined targets of PL emission maximum, which also correlate to specific scattering maxima. In this way, the maximum emission wavelength of all NPs could be tuned with an accuracy and precision within 2–3 nm. This procedure allows tailored fabrication of arrays of supported Au NPs with precisely defined positions and sizes.

2. Results and Discussion

The experimental setup employed in this work is presented in Figure 1a. A circularly polarized 532 nm continuous wave laser is incorporated into a dark-field upright microscope and focused at the sample plane with a 1.0 NA water immersion objective (beam waist 342 ± 5 nm). In addition to dark-field imaging, the microscope is coupled to a spectrometer, allowing the acquisition of scattering and PL spectra of individual NPs. The elastically scattered light from the laser is reflected using a notch-filter toward a confocal detection channel. Scanning and sample positioning is achieved using a piezoelectric stage.

Plasmon-driven growth was studied on single, spherical Au NPs, 66 nm in diameter, supported on glass substrates.

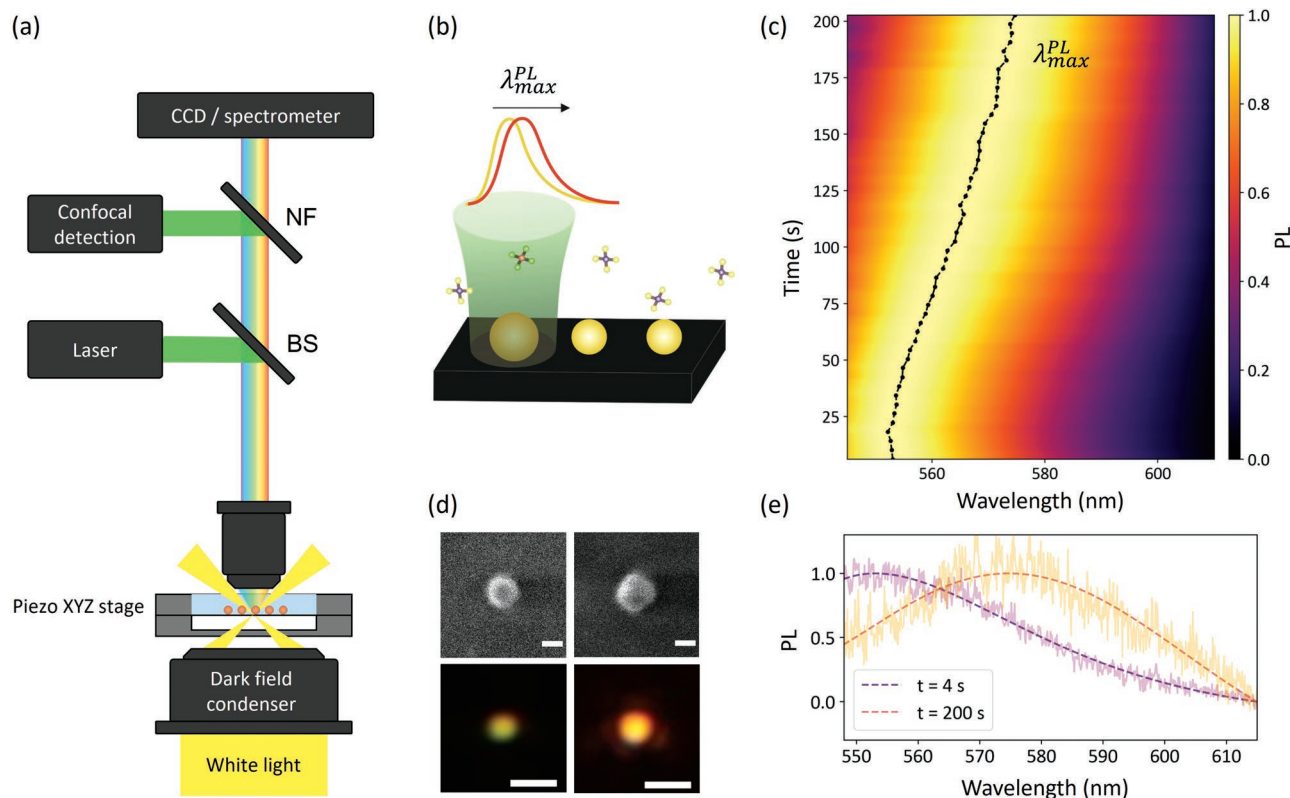


Figure 1. a) Scheme of the experimental set-up. BS: beam-splitter. NF: notch filter. b) Schematic illustration of the plasmon-driven NP growth: 66 nm Au NPs optically printed on a glass substrate are put in contact with an aqueous solution of HAuCl₄ and irradiated individually with a 532 nm CW laser for a given time interval. During the growth, PL spectra are acquired periodically, and the wavelength of maximum PL emission ($\lambda_{\max}^{\text{PL}}$) is followed in time. c) Example PL spectra (normalized) of an Au NP, acquired every 4 s of integration time during 200 s of a grow reaction with laser irradiance of $2.6 \text{ mW } \mu\text{m}^{-2}$ in the presence of $0.5 \text{ } \mu\text{M}$ HAuCl₄ (aq). The black line indicates $\lambda_{\max}^{\text{PL}}$. d) Scanning electron microscopy images (upper panels, scale bar = 50 nm) and dark-field images (lower panels, scale bar = $1 \text{ } \mu\text{m}$) of the example NP before (left) and after the growth reaction (right). e) Comparison between initial and final PL spectra, for an irradiation time of 200 s. The solid lines are the raw signals and the dashed lines their fittings with a polynomial function.

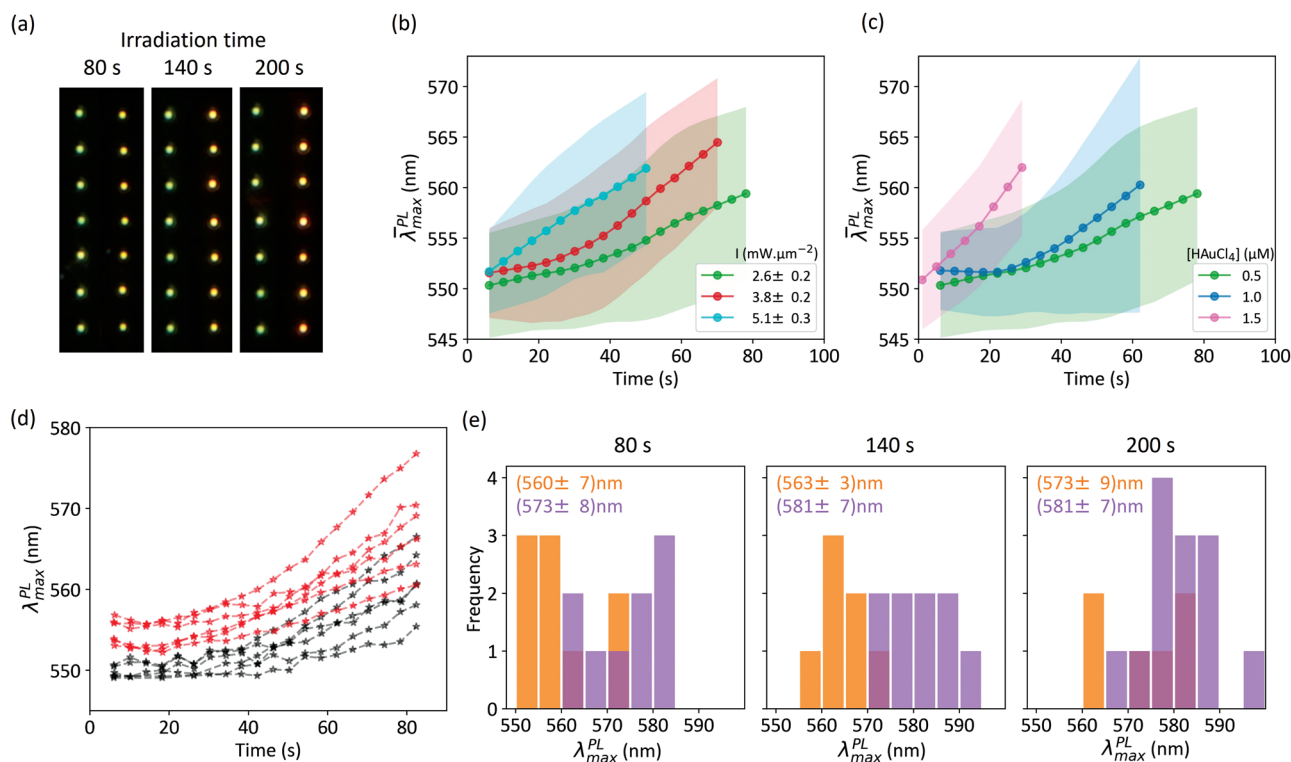


Figure 2. Kinetics of the growth reaction. a) Dark-field image of the same NPs before (left) and after (right) irradiation during different time lapses: 80, 140, and 200 s. The interparticle distance between NPs of the same column is $3 \mu\text{m}$. b) Mean $\lambda_{\max}^{\text{PL}}$ versus time during plasmon-driven growth at three different laser irradiances. Each color curve corresponds to the average kinetics of 10 NPs irradiated in the presence of $0.5 \mu\text{M}$ HAuCl₄ (aq). Shaded areas indicate ± 2 standard deviations. c) Mean $\lambda_{\max}^{\text{PL}}$ versus time for plasmon-driven growth at different HAuCl₄ concentration. Each color curve corresponds to the average kinetics of 14 NPs irradiated at a fixed irradiance of $2.6 \text{ mW} \mu\text{m}^{-2}$. Shaded areas indicate ± 2 standard deviations. d) Maximum of PL emission $\lambda_{\max}^{\text{PL}}$ versus time for 12 NPs grown for 80 s under identical conditions: irradiance = $2.6 \text{ mW} \mu\text{m}^{-2}$, $0.5 \mu\text{M}$ HAuCl₄. Two sets of NPs are shown. In black, 6 NPs with initial $\lambda_{\max}^{\text{PL}}$ between 549 and 551 nm. In red, 6 NPs with initial $\lambda_{\max}^{\text{PL}}$ between 553 and 557 nm. e) Histograms of the final $\lambda_{\max}^{\text{PL}}$ of NPs measured on water after plasmon-driven growth for three irradiation times (80, 140, and 200 s) with $0.5 \mu\text{M}$ HAuCl₄ and laser irradiance of $2.6 \text{ mW} \mu\text{m}^{-2}$. The orange and violet colors correspond to two replications of the experiment. The mean and standard deviation of the distributions are indicated.

Their size distribution is shown in Figure S1, Supporting Information. The process is illustrated in Figure 1b. An NP is positioned in the center of the laser beam and irradiated in the presence of HAuCl₄ aqueous solution. The laser excites the plasmon resonance, driving the local reduction of gold precursor on the NP surface and leading to NP growth.^[42] Simultaneously, while the reaction occurs, the PL emission excited with the same laser is measured. The growth of the NP leads to a redshift in the PL emission, allowing the live monitoring of the process. PL spectra is acquired in intervals of 4 s (integration time), and the peak PL emission wavelength ($\lambda_{\max}^{\text{PL}}$) is determined from a fit of the spectrum to a polynomial function. Figure 1c shows an example of the normalized PL emission spectra of a single NP versus time during a growing process over 200 s. The dashed black line indicates $\lambda_{\max}^{\text{PL}}$. Figure 1d illustrates the size increase (scanning electron micrographs, upper panels) and change of optical properties (dark-field images, lower panels) undergone by this particular NP after 200 s of growth. The PL spectrum of the same NP before and after being irradiated for 200 s is shown in Figure 1e. The error in the determination of $\lambda_{\max}^{\text{PL}}$ was estimated by repeatedly measuring the PL signal and resulted in 1 nm (Figure S2, Supporting Information).

As reported, NP growth can be controlled by managing the light dose and reaction conditions.^[39,40,42] This is illustrated by the experiments shown in Figure 2a–c. Arrays of isolated Au NPs were prepared on a glass substrate by optical printing and subsequently grown under different conditions. Figure 2a shows dark-field images of arrays of NPs before and after 532 nm irradiation ($2.6 \text{ mW} \mu\text{m}^{-2}$) in the presence of $0.5 \mu\text{M}$ HAuCl₄ (aq) for three different irradiation times: 80, 140, and 200 s. Repetition of such experiments enables the determination of average growth kinetics, in terms of $\lambda_{\max}^{\text{PL}}$ as a function of time, for different conditions of laser irradiance and concentrations of HAuCl₄, as shown in Figure 2b,c. The shaded area around the average kinetics marks two standard deviations. Clearly, knowing the average growth kinetics only provides a limited degree of control and predictability over the final size and properties of the NPs due to significant variations from one NP to another. Figure 2d shows the growth kinetics for 12 nominally identical NPs grown under the same reaction conditions. The first observation is that the initial $\lambda_{\max}^{\text{PL}}$ has a variability from 549 to 557 nm (the notch filter impedes the measurements below 549 nm), which can be ascribed to the original size distribution of the NPs. This initial dispersion of sizes and optical properties is typically increased during growth due to variability

in the growth rate of the individual NPs. To illustrate this effect, the 12 NP growth trajectories of Figure 2d are separated in two sets with different spread of initial $\lambda_{\text{max}}^{\text{PL}}$ (more details in Figure S3, Supporting Information). The differences between individual kinetics could be caused by different factors. First, the actual plasmon excitation of each NP may vary due to: i) slight misalignments of the NP position with respect to the focused laser beam during the experiment (the lateral drift of our set-up was determined to be of 0.45 nm s^{-1} , which may cause irradiance variations up to 15% in 200 s), and ii) the absorption and scattering cross section of each NP may present variations due the initially different sizes (see size distribution on Figure S1, Supporting Information and the scattering spectra on Figure S4, Supporting Information). The variations in the plasmon excitation have a direct effect on the growth reaction kinetics, and also lead to a different temperature increase for each NP.^[50] The latter, in turn, may affect the growing kinetics directly or indirectly due to modifications of the local concentration of reagents product of the thermal gradients. Second, the local concentration of reagents may also vary from one NP to another due to variations in the local surface charge of the substrate or the differently sized NPs under illumination.^[51–53] Third, for this particular reaction the age of the gold precursor is also an important parameter,^[54] further compromising the reproducibility of the growing kinetics.

The lack of reproducibility in the final properties of the NPs is further illustrated by the histograms of final $\lambda_{\text{max}}^{\text{PL}}$ shown in Figure 2e. For three different irradiation times (80, 140, and

200 s), two sets of nominally identical experiments (indicated as orange and violet bars) deliver statistically different distributions of $\lambda_{\text{max}}^{\text{PL}}$.

The previous analysis clearly shows the limitations of tuning the optical properties of NPs by simply controlling the irradiation time and reaction conditions. Therefore, we investigated a strategy of plasmon-driven NP growth with a closed-loop control, as schematically shown in Figure 3a. The Stokes-shifted PL spectrum of each NP is monitored while it grows, and its maximum is determined in real time. When $\lambda_{\text{max}}^{\text{PL}}$ reaches a target value, the reaction is stopped by automatically switching off the illumination. Figure 3b shows $\lambda_{\text{max}}^{\text{PL}}$ versus time for the growth of 12 individual NPs with a target of $\lambda_{\text{max}}^{\text{PL}} = 570 \text{ nm}$, indicated with a dashed red horizontal line. Although the initial $\lambda_{\text{max}}^{\text{PL}}$ and the growing kinetics vary from case to case, the final $\lambda_{\text{max}}^{\text{PL}}$ of all NPs is highly reproducible. The standard deviation of the final $\lambda_{\text{max}}^{\text{PL}}$ is less than 1 nm, as indicated with the shaded grey area. The final mean $\bar{\lambda}_{\text{max}}^{\text{PL}}$ is indicated with a dashed horizontal black line, showing small redshift with respect to the target in the order of 1 nm, corresponding to further growing of the NPs during the time-lapse associated with the automated stopping of the reaction. Subsequently, the HAuCl_4 solution was removed, the substrate was thoroughly rinsed, and the PL spectra were acquired in water environment. The final homogeneity in the optical properties is also evident in the dark-field images of different NPs before (green) and after (orange) growing, as shown in Figure 3c. The histogram of the $\lambda_{\text{max}}^{\text{PL}}$ for the NPs grown to three different targets (560, 570, and 580 nm)

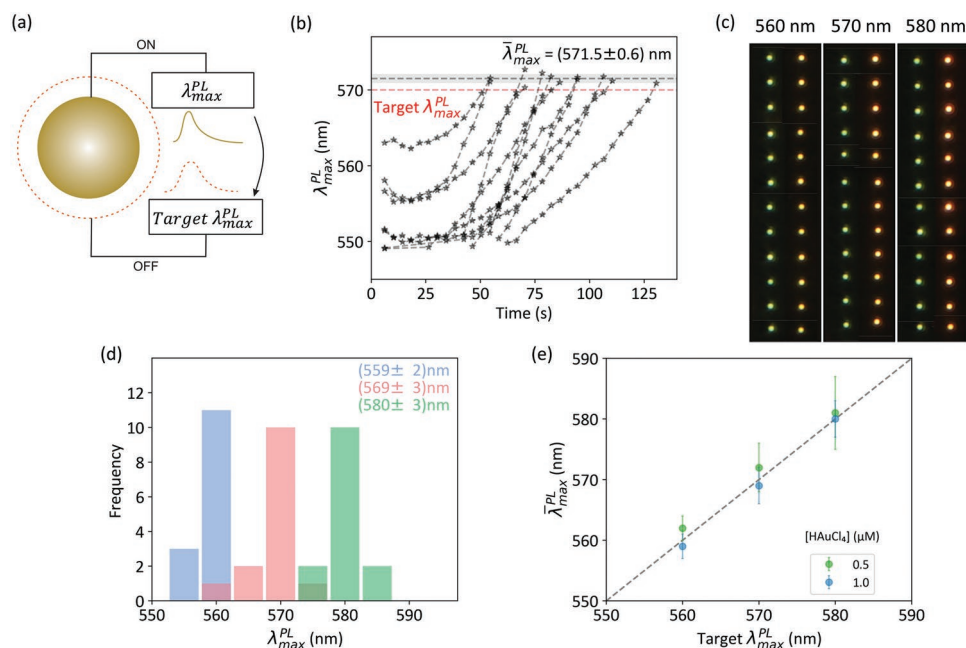


Figure 3. a) Scheme of the closed-loop plasmon-driven NP growth to fine-tune the optical properties of single Au NP. The PL spectra are monitored during the irradiation, which is stopped when $\lambda_{\text{max}}^{\text{PL}}$ reaches a predefined target value. b) $\lambda_{\text{max}}^{\text{PL}}$ versus time for the growing of 12 individual NPs, with a target of $\lambda_{\text{max}}^{\text{PL}} = 570 \text{ nm}$, indicated with a dashed red horizontal line. The dashed black line indicates the final mean $\bar{\lambda}_{\text{max}}^{\text{PL}}$, and the grey shaded area corresponds to ± 1 standard deviation. c) Dark-field images of the same Au NPs before and after irradiation for different $\lambda_{\text{max}}^{\text{PL}}$ targets: 560, 570, and 580 nm. The interparticle distance between NPs of the same column is $3 \mu\text{m}$. d) Histogram of the final $\lambda_{\text{max}}^{\text{PL}}$ measured in water (no HAuCl_4) for the three different targets: 560 nm (in blue), 570 nm (in red) and 580 nm (in green). The concentration was $1 \mu\text{M}$ HAuCl_4 . The labels show the mean and the standard deviation for each condition. e) Mean of the final $\bar{\lambda}_{\text{max}}^{\text{PL}}$ versus target $\lambda_{\text{max}}^{\text{PL}}$. The grey dotted curve indicates the identity. The green data points correspond to a concentration of $0.5 \mu\text{M}$ HAuCl_4 and blue data points to $1 \mu\text{M}$ HAuCl_4 . The laser irradiance for all experiments presented in this figure was $2.6 \text{ mW } \mu\text{m}^{-2}$.

after the washing step is shown in Figure 3d. In all cases the $\lambda_{\max}^{\text{PL}}$ matches the target with a precision and an accuracy of $\approx 2\text{--}3$ nm. This slightly larger standard deviation compared to the one obtained while measuring online could be caused by a small modification of the NPs or their local environment after stopping the reaction and rinsing. The correlation between the target and the obtained $\lambda_{\max}^{\text{PL}}$ is shown in Figure 3e for two different growing experiments using HAuCl_4 concentration of either $0.5\ \mu\text{M}$ (green) or $1\ \mu\text{M}$ (blue). Reproducible growth is demonstrated, even with a factor 2 change in the HAuCl_4 concentration.

The results shown so far demonstrate that the closed-loop control strategy allows the reproducible fine tuning of the PL of individual NPs. To study how this is translated to other properties, the final size and scattering spectra of the NPs were analyzed. **Figure 4a** shows the PL emission spectrum (solid line), the scattering spectrum (dotted line), and a scanning electron microscopy image (right panel) for four example NPs: an original seed NP, and three NPs grown in closed-loop to $\lambda_{\max}^{\text{PL}}$ targets of 560, 570, and 580 nm. In all cases, the growth

was performed with an irradiance of $2.6\ \text{mW}\ \mu\text{m}^{-2}$ and $0.5\ \mu\text{M}$ HAuCl_4 .

Figure 4b shows histograms of the maximum scattering wavelengths $\lambda_{\max}^{\text{S}}$. In orange, the histogram corresponding to the optically printed Au seed NPs before growth is presented, showing a standard deviation of 3 nm. The same NPs, after growing to a $\lambda_{\max}^{\text{PL}}$ target of 560 nm, present a similarly narrow distribution, as shown in the blue histogram. (Histograms of $\lambda_{\max}^{\text{S}}$ for NPs grown to the other targets are shown in Section S5, Supporting Information). On the other hand, if the NPs are grown for a given reaction time, the distribution widens to a standard deviation of 11 nm, as shown in the grey histogram. These results indicate that the closed-loop growth based on PL emission also provides control over the scattering properties. Figure 4c (grey dots) shows a correlation scatter plot between $\lambda_{\max}^{\text{PL}}$ and $\lambda_{\max}^{\text{S}}$ for 198 Au NPs, including all NPs studied in this work, that is, NPs grown by controlling the reaction time or using the closed-loop to a specific target. 95% of the NPs present a blueshift of $\lambda_{\max}^{\text{PL}}$ relative to $\lambda_{\max}^{\text{S}}$. This observation is in agreement with previous reports on Au nanorods,^[55]

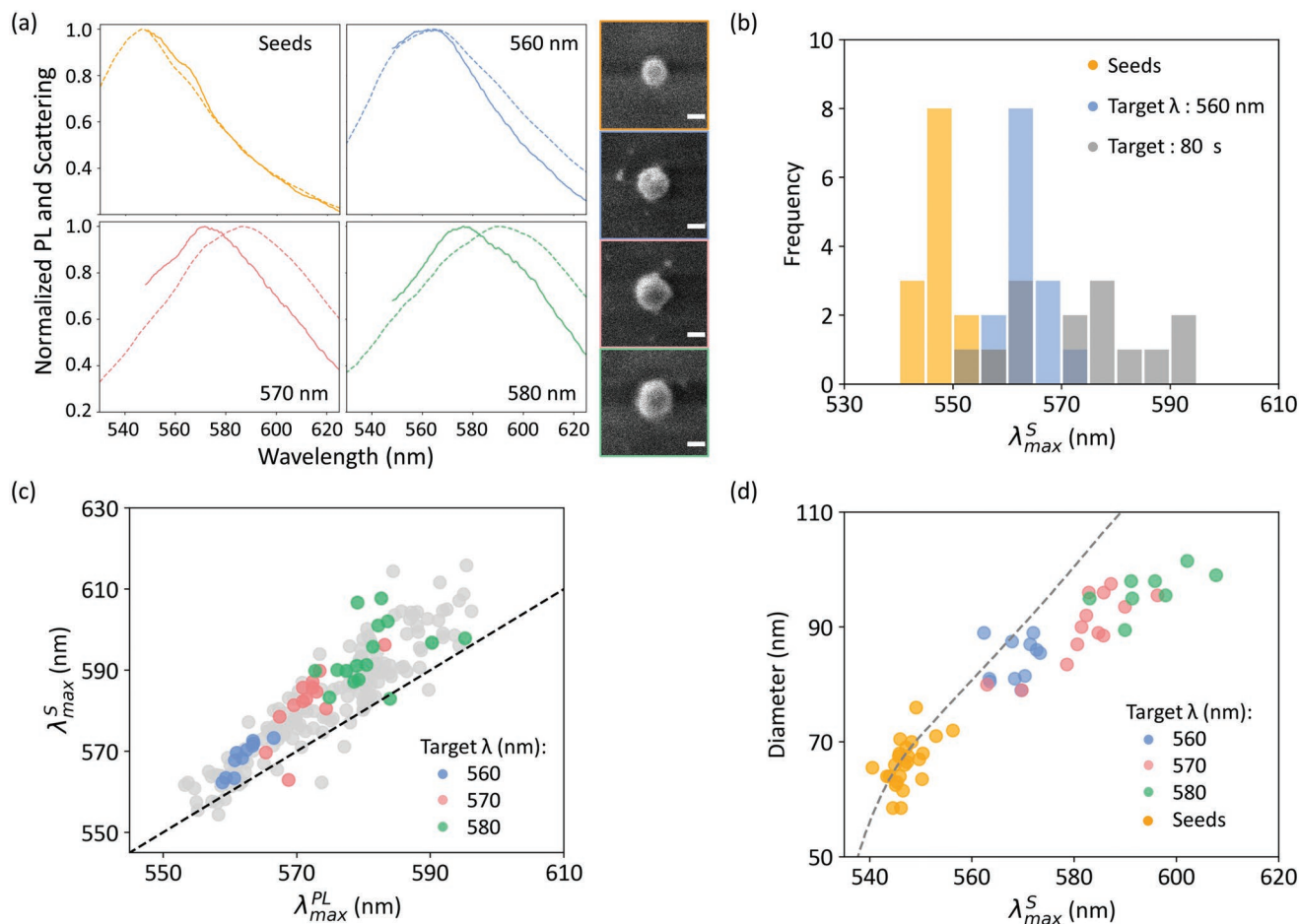


Figure 4. a) PL Stokes spectra (solid line) and scattering spectra (dotted line) of four exemplary Au NPs: a seed NP before plasmon-driven growth (orange color), and NPs grown with $\lambda_{\max}^{\text{PL}}$ targets of 560 (blue color), 570 (red color), and 580 nm (green color). Their scanning electron microscopy images are shown in the right panels (scale bar = 50 nm). b) Histograms of the $\lambda_{\max}^{\text{S}}$ for the seed NPs (orange; $\lambda_{\max}^{\text{S}} = 547 \pm 3$ nm), for NPs grown with a $\lambda_{\max}^{\text{PL}}$ target of 560 nm (blue; $\lambda_{\max}^{\text{S}} = 564 \pm 3$ nm), and for NPs grown for a fixed irradiation time of 80 s (grey; $\lambda_{\max}^{\text{S}} = 573 \pm 11$ nm). $N = 14$ NPs for each distribution. c) Final $\lambda_{\max}^{\text{S}}$ versus final $\lambda_{\max}^{\text{PL}}$ for 198 grown Au NPs. The color dots represent the Au NPs grown in closed-loop to the three $\lambda_{\max}^{\text{PL}}$ targets. The black dashed line represents the identity. d) Diameter (nm) versus $\lambda_{\max}^{\text{S}}$ for the seed NPs and for the NPs grown to the three $\lambda_{\max}^{\text{PL}}$ targets. The grey dotted line is the curve predicted by the Mie theory of Au spherical NPs on water.

nanodiscs,^[56] and spheres.^[47,57,58] Applying the controlled growth based on $\lambda_{\text{max}}^{\text{PL}}$ to reach a precise target of $\lambda_{\text{max}}^{\text{S}}$ would require a deeper understanding or characterization of the relationship between scattering and PL. Nonetheless, the controlled growth targeting three different $\lambda_{\text{max}}^{\text{PL}}$ led to three populations of NPs with well distinct $\lambda_{\text{max}}^{\text{S}}$ (Figure 4c, colored data points). Finally, a correlation scatter plot of diameter versus $\lambda_{\text{max}}^{\text{S}}$ is presented in Figure 4d for the seed NPs, and for the NPs grown to the three $\lambda_{\text{max}}^{\text{PL}}$ targets. Again, three populations are observed, meaning that tuning the PL emission spectra also translates into different NPs sizes. The gray dotted line represents the Mie theory for spherical Au NP on water. In a previous work, we showed that the diameter and scattering for spherical Au NPs printed on glass followed the Mie theory as if the Au NPs were embedded only on water, with little influence of the substrate.^[50] However, it can be observed that the grown Au NPs do not follow the Mie theory, presenting a larger $\lambda_{\text{max}}^{\text{S}}$. One possible reason for this deviation could be that the grown Au NPs are not perfectly spherical, generating a larger contact area with glass, and red shifting the spectra due to the refractive index increase.

3. Conclusions

We have shown the finely tuned growth of single supported Au spherical NPs by online monitoring the advance of a plasmon-driven reduction of aqueous HAuCl₄. The growth is monitored non-invasively by registering the PL emission spectrum from the NPs excited with the same laser used to induce the plasmon-driven reaction. When a target wavelength of maximum PL is reached, the growth reaction is stopped by switching off the plasmon excitation. In this way, the PL emission maximum of single NPs could be tuned between 550 and 580 nm with a precision of 2–3 nm. The tuning was also translated in other properties as the final size and wavelength of maximum scattering of the NPs. In agreement to previous reports, we observed that the PL emission spectrum is blue-shifted with respect to the scattering. Thus, using the closed-loop control based on PL to attain specific scattering properties would require a detailed characterization of the blue-shift.

In comparison to NP growth controlled by reaction time and/or conditions, the closed-loop control avoids the effect of varying reaction rates due to different NP size, local irradiance, and/or local concentrations of reagents, leading to a three to fourfold improvement in the reproducibility of the final properties of the NPs.

This work introduces a platform for the fabrication of single supported NPs with high control of their final optical properties. Importantly, the method is not limited to the spectral range investigated in this work and is easily adaptable to other plasmonic NPs. For example, other metal precursors such as H₂PtCl₆^[37,44] may be used. Therefore, its application for the fabrication of hybrid systems like bimetallic core@shell NPs with precisely defined sizes and compositions is envisioned. Another potential application would be the fine tuning of plasmonic resonators or optical nanoantennas. For example, NP dimer resonators could be tuned with high precision by growing the NPs and monitoring the coupled resonance which is highly sensitive to the gap between the NPs.^[26,59] Finally, the closed-loop control could also be used in the absence of any

precursor to control the photothermal reshaping, sintering, or melting of NPs, for instance to fabricate alloyed NPs from melting bimetallic core@shell NPs.

In summary, the PL closed-loop control of plasmon driven reactions on NPs offers a clear route for the fabrication of supported arrays of NPs with individually tailored optical properties and functionalities, relevant to numerous applications and scientific investigations.

4. Experimental Section

Gold Nanoparticles and Precursor: Negatively charged Au NPs were purchased from Nanopartz (NPC functionalized, SEM measured diameter = 66 ± 4 nm). To adjust the concentration for optical printing and eliminate possible aggregates, the purchased colloidal suspension (OD = 1) was diluted 4 times in a 1.5 mM NaCl solution and centrifuged at 600 RCF for 5 min (the supernatant was used). 99.9% HAuCl₄ was purchased from Sigma Aldrich. A 0.01 M stock solution was prepared in ultrapure water and stored at 4 °C.

Substrate Functionalization: Polydiallyldimethylammonium chloride (PDPA, MW ≈ 400 000–500 000) and sodium polystyrene sulfonate (PSS, MW = 70 000) were purchased from Sigma Aldrich and used without further purification. Soda-lime glass (Paul Marienfeld GmbH & Co, DE) substrates were cleaned using Hellmanex III 0.2% for 10 min in an ultrasound bath at room-temperature. After rinsing with deionized water and acetone, they were dried in an oven at 85 °C for 2 h. Their surface was plasma-activated using a plasma cleaner (Diener, Zepto) at 75 W and 0.5 mbar for 3 min. Finally, the glass substrates were rendered with negative surface charge by sequentially immersing them in a PDPA solution (1 mg mL⁻¹ in 0.5 M NaCl) for 15 min, copious rinsing with ultrapure water, immersion in a PSS solution (1 mg mL⁻¹ in 0.5 M NaCl) for 15 min, and rinsed again with ultrapure water.

Optical Printing and Plasmon-Driven Nanoparticle Growth: Au NPs were optically printed onto each substrate according to the process described in the previous works.^[29,30] Briefly, the Au NP suspension was placed on top of the functionalized substrate having the same charge as the NPs in an open chamber and printed using a CW 532 nm focused laser (Ventus, Laser Quantum) using a 60× water-immersion (Olympus) objective with a NA 1 (beam waist $\omega_0 = (342 \pm 5)$ nm). Printing irradiance was adjusted to 7 mW μm^{-2} . After the printing process, the colloidal suspension was removed, and the chamber was rinsed several times with ultrapure water.

For the plasmon-driven growth, the water was replaced by a freshly prepared HAuCl₄ (concentrations between 0.5 and 1.5 μM). Laser irradiances were kept between 2.6 and 5.1 mW μm^{-2} depending on each experiment.

Dark-field images were acquired using a digital color camera (Canon EOS-500D). Single NP scattering spectra were obtained with a spectrometer (Shamrock 500i, Andor) based on 150 lines mm⁻¹ grating and an EMCCD camera (Andor Ixon EM + 885). The same camera and spectrometer were employed to measure online the Stokes PL emission. Scanning electron microscopy images were obtained with a Quanta FEG 250. The entire process of optical printing, online spectral acquisition, and analysis was fully automated by Python routines.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

nanoparticle arrays, optical printing, plasmonic chemistry, single-particle spectroscopy

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